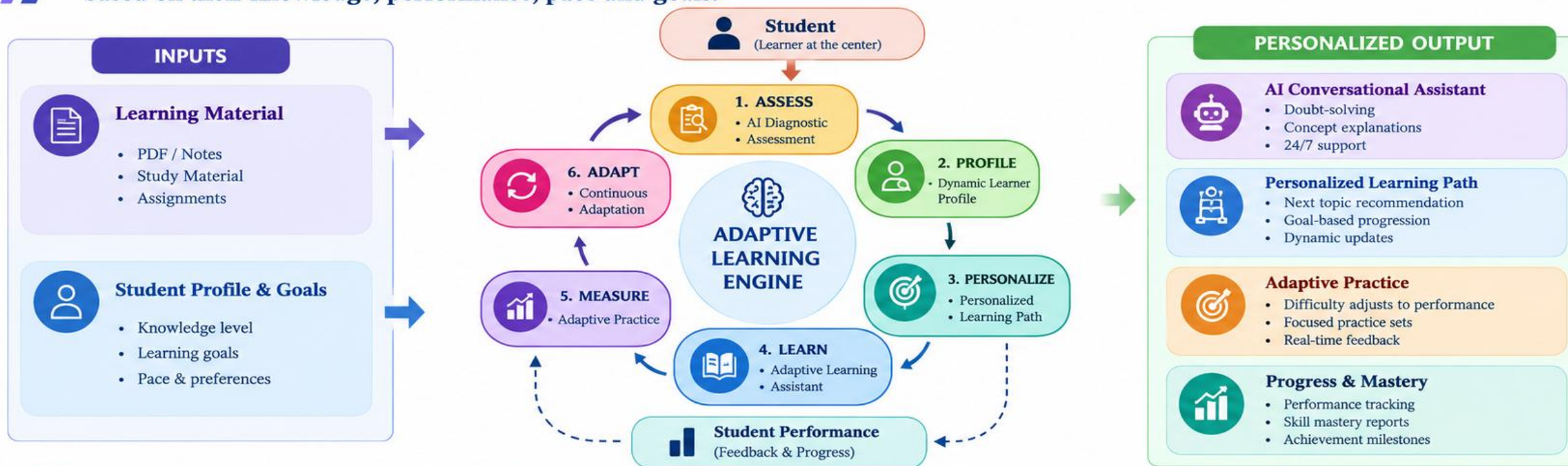


ADAPTED AI

Personalized Learning That Adapts to Every Student

- **Theme** - Smart Education.
- **Problem Statement - PS1** - Personalized AI Learning Assistant.
- **Team Name** – SkillMatix.
- **Team members name** –
 1. Rutuja Deshmukh
 2. Rajshree Patil
 3. Manthan Hingmire.
 4. Prathamesh Gadakh.
- **College name** - MET Bhujbal Knowledge City, Institute of Engineering.

An AI-powered adaptive learning platform that continuously personalizes a student's learning journey based on their knowledge, performance, pace and goals.



How It Addresses the Problem

The Problem

The Solution

	Different knowledge levels	→		Personalized learning paths
	Unknown learning gaps	→		AI diagnostic + topic mastery analysis
	Static study plans	→		Continuously adapting learning paths
	Same difficulty for everyone	→		Performance-based adaptive practice
	Generic learning support	→		Personalized Adaptive Learning Assistant



Innovation & Uniqueness

Continuous Adaptive Learning Loop



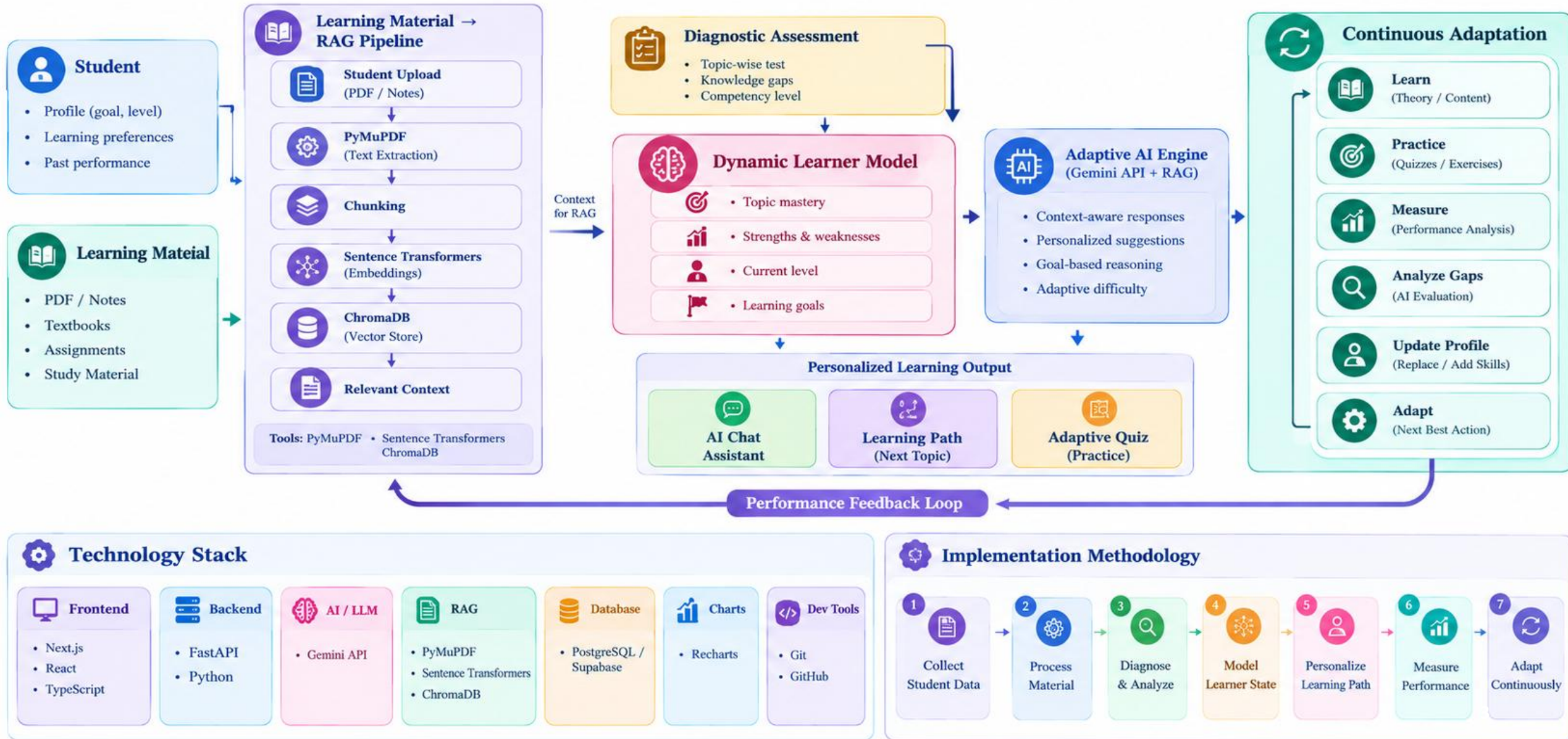
Instead of generating a personalized study plan only once, AdaptEd AI continuously updates the learner profile and changes the next learning action as the student progresses.

- Student-centric approach
- Understands the learner first
- Dynamic learner profile
- Personalized learning path
- AI conversational assistant
- Practice difficulty responds to performance
- Continuous updates from performance
- Path changes with new evidence



Not a fixed study plan — a learning journey that adapts to you.

Full-stack AI + RAG + Adaptive Learning Architecture



Practical to build, scalable to deploy, and designed for real student learning needs.



Feasibility



Technical Feasibility

- Uses mature, accessible technologies: Next.js, FastAPI, Python, Gemini API, RAG, ChromaDB, PostgreSQL.

Modular architecture allows each component to be developed and tested independently.



Cost Feasibility

- Low infrastructure requirement for MVP
 - API-based AI
 - Open-source RAG components
 - Lightweight database
 - Cloud-ready architecture



Low-cost prototype using available APIs and open-source tools.



Scalability

- Modular architecture supports future expansion.

Future possibilities:

- More subjects
- More students
- Multilingual learning
- Mobile application
- Institutional deployment
- Advanced analytics



Implementation Feasibility

- Focused MVP can be developed within the hackathon using modular components.



Challenges & Risks

1



RAG Accuracy

Risk:

Retrieved content may be incomplete or irrelevant.

2



AI Hallucination

Risk:

LLM may generate incorrect or unsupported information.

3



API Dependency

Risk:

AI API limits, latency or connectivity issues can affect responses.

4



Student Data Privacy

Risk:

Uploaded learning materials and learner performance data need protection.

5



Limited Initial Data

Risk:

May affect accuracy of learner modeling in early stages.



Mitigation Strategies



RAG Validation

→ Better grounding and relevant context.



Minimal Data Collection

→ Privacy and security by design.



Feedback + Evaluation

→ Quality control and continuous improvement).



Viability



Low infrastructure



Modular



Scalable

Transforming one-size-fits-all learning into continuous, personalized learning.



Student Impact

- Personalized Learning**
 - Learning path based on individual goals
 - Content adapted to current level
- Better Gap Identification**
 - Detects weak topics & prerequisite gaps
 - Shows what to improve
- Efficient Learning**
 - Reduces time on mastered topics
 - Focuses practice where needed
- Continuous Improvement**
 - Performance updates learner profile
 - Learning path evolves over time
- Greater Learning Awareness**
 - Understand strengths, weaknesses & progress

Educator Benefits

- Learner Insights**
 - Track progress & performance
 - Identify struggling students
- Targeted Support**
 - Personalized intervention strategies
 - Save time on manual monitoring
- Better Decision Making**
 - Data-driven teaching strategies
 - Measure learning outcomes
- Enhanced Engagement**
 - More interactive learning
 - Improved student participation

Institution Impact

- Learning Analytics**
 - Track overall performance
 - Improve academic planning
- Scalable Learning**
 - Serve more students effectively
 - Flexible deployment
- Better Resource Use**
 - Optimize content and faculty effort
 - Reduce drop-off rates
- Early Intervention**
 - Detect at-risk students
 - Improve retention

The Bigger Picture

- More equitable access to quality education
- Reduced learning gaps across all backgrounds
- A smarter, more inclusive and future-ready education system

Broader Impact

- Social**: More equal opportunities for every learner
- Economic**: Better skill development and workforce readiness
- Educational**: Stronger learning ecosystems and higher achievement

Future Scope

Multilingual AI > Mobile App > Institutional Analytics

Advanced Learner Modeling > Offline / Low-connectivity Support

Research-backed design for responsible, personalized and *adaptive learning*



RESEARCH FOUNDATION

Our research focuses on five foundations:

- 1 Adaptive Learning**
Adaptive e-learning adapts the learning experience to individual learners instead of a one-size-fits-all approach.
- 2 Learner Modelling**
A learner model captures knowledge, previous performance and learning activities to support personalization.
- 3 Adaptive Assessment**
Adaptive assessment helps improve personalization and identify knowledge gaps.
- 4 AI-Powered Personalization**
AI techniques support learner classification, performance prediction and dynamic content sequencing.
- 5 Responsible GenAI**
UNESCO's guidance emphasizes human-centred, safe, equitable and privacy-conscious use of GenAI in education.



KEY REFERENCES

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ScienceDirect (2024)
- 2 Learner modelling in adaptive e-learning**
DOI (2024)
- 3 Adaptive assessment and personalization**
ScienceDirect (2024)
- 4 AI in adaptive education: techniques for personalized learning**
Springer (2025)
- 5 Guidance for generative AI in education and research**
UNESCO (2023)



RESEARCH



DESIGN



ADAPT



IMPROVE



CONCLUSION

Same subject $\neq$ Same learner $\neq$ Same learning path

AdaptEd AI combines learner modelling, RAG, AI assistance and continuous performance feedback to create a learning journey that adapts to the individual student.

LEARN $\rightarrow$ MEASURE $\rightarrow$
ADAPT $\rightarrow$ IMPROVE